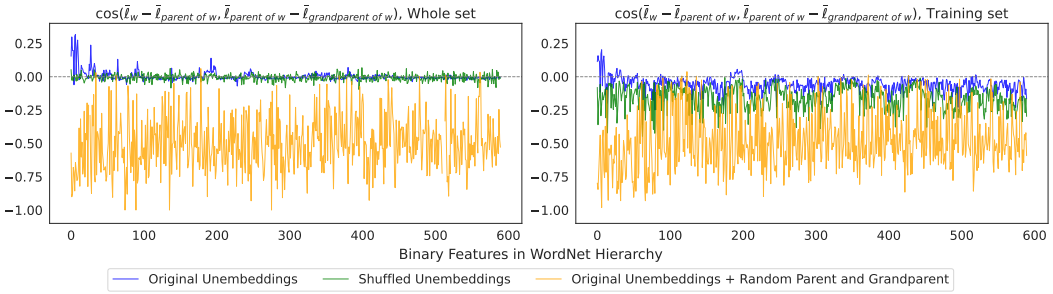

Figure 10: WordNet noun hierarchy is encoded as the orthogonal structure predicted by statement (d) in Theorem 8. The cosine similarity between a child-parent vector and a parent-grandparent vector for each feature in the hierarchy estimated by original (blue) and shuffled (green) unembedding vectors with the whole (left) and training (right) set of tokens. Another baseline (orange) is the cosine similarity with random parent and grandparent vectors.

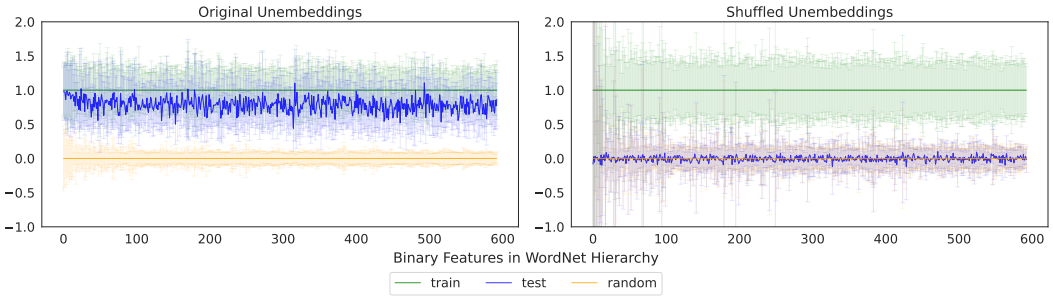

Figure 11: Comparison of projection of train (green), test (blue), and random (orange) words on estimated **mean vector** for each WordNet feature, from the original unembeddings (left) and the shuffled unembeddings (right). The value for each word y is $(g(y)^\top \bar{\ell}_w)/\|\bar{\ell}_w\|_2^2$ that we expect to be 1 when y has the target feature. The x-axis indices denote all features in the noun hierarchy. The thick lines present the mean of the projections for each feature and the error bars indicate the standard deviation. The features are ordered by the hierarchy.

Similar to the findings in Figure 5, the shuffled unembedding vectors (green) reflect the set inclusion relationships, and the cosine similarities are close to 0. When we violate the set inclusion by using only the 70% training set of tokens, the right plot in Figure 10 shows that the cosine similarities from shuffled (green) unembeddings are much smaller, while those from original (blue) unembeddings are still close to 0. This indicates that the hierarchical orthogonality is not a trivial consequence of the set inclusion, but rather reflects the semantic hierarchy.

The Mean Estimator for the Vector Representation Has High Variance The mean vector $\mathbb{E}(g_w)$ can serve as an estimator for the vector representation $\bar{\ell}_w$. However, Figure 11, which is analogous to Figure 3, shows that the projections of train and test words onto the mean vector have larger variances. Therefore, the LDA direction (eq. (5.2)) would be a more appropriate estimator for the vector representation.

Geometry of Hierarchical relations on the Euclidean Inner Product In this paper, we transform the representation spaces to use the causal inner product. However, what if we employ the naive Euclidean inner product instead? To address this, we estimate the vector representations for the WordNet noun hierarchy estimated from the original Gemma unembedding vectors through centering alone, without applying whitening (which is necessary for the causal inner product). This approach still preserves the Euclidean inner product. Then, Figure 12 shows that the hierarchical orthogonality is not satisfied in the Euclidean inner product space, as evidenced by the cosine similarities not being close to 0.

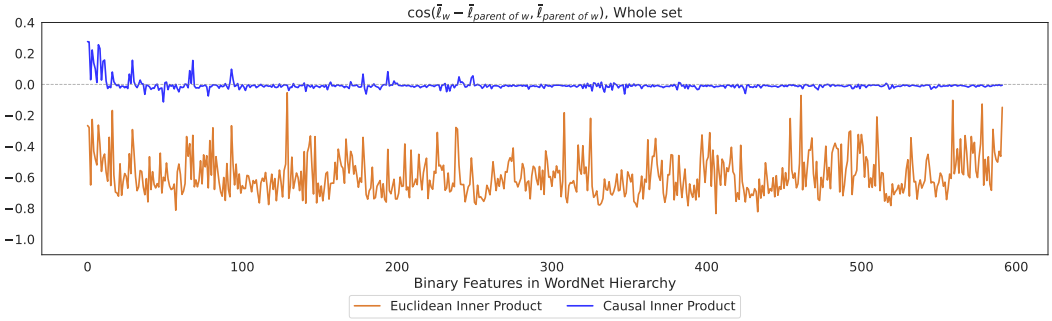

Figure 12: WordNet noun hierarchy is not encoded as the orthogonal structure when we use the naive Euclidean inner product, whereas it is encoded as orthogonality in the causal inner product.

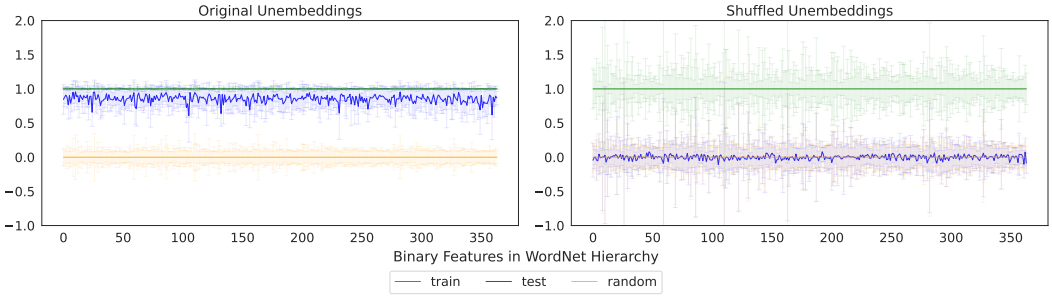

Figure 13: Vector representations exist for most binary features in the WordNet **verb** hierarchy.

WordNet Verb Hierarchy In the same way as for the noun hierarchy, we estimate the vector representations for the WordNet verb hierarchy. Analogous to Figure 3, Figure 13 shows that the vector representations exist for most binary features in the WordNet verb hierarchy. Analogous to Figure 4, Figure 14 shows that the hierarchical semantics in WordNet are encoded in the Gemma representation space. Analogous to Figure 5 and Figure 10, Figure 15 and Figure 16 show that the hierarchical orthogonality is encoded in the Gemma representation space for the WordNet verb hierarchy.

WordNet Noun Hierarchy encoded in LLaMA-3 Model We also validate the theoretical predictions in the LLaMA-3-8B model (Dubey et al., 2024) in the same way as for the Gemma model. The vector representations for most binary features in the WordNet noun hierarchy exist in the LLaMA-3 model (Figure 17). The hierarchical semantics in WordNet are encoded in the LLaMA-3 representation space (Figure 18). Lastly, the hierarchical orthogonality is encoded in the LLaMA-3 representation space for the WordNet noun hierarchy (Figure 19 and Figure 20).

G WHY DOES SET INCLUSION GIVE THE ORTHOGONALITY?

The left panel of Figure 5 shows that the cosine similarity between the child-parent vector and the parent vector is also close to 0 when they are estimated by the shuffled unembedding vectors. One of the possible explanations for this phenomenon is that the set inclusion relationships (and the estimating process) give the orthogonality. For example, the set inclusion between the collections $\mathcal{Y}(z) = \{\texttt{dog}, \texttt{sandwich}, \texttt{running}\}$ for a child z and $\mathcal{Y}(w) = \{\texttt{dog}, \texttt{sandwich}, \texttt{running}, \texttt{France}, \texttt{scientist}, \texttt{diamond}\}$ for a parent w can make the child-parent vector and the parent vector orthogonal.

Since the LDA-based vector representation estimated by eq. (5.2) is similar to the mean vector $\mathbb{E}(g_w)$, we briefly derive the orthogonality using the mean vector as the vector representation. Suppose that $a_1, \ldots, a_{N_a}, b_1, \ldots, b_{N_b} \overset{i.i.d.}{\sim} \mathcal{N}(0, I_d)$, which may correspond to the (centered and whitened) shuffled unembeddings. Here, a_i's are for the child z, and both a_i's and b_i's are for the par-

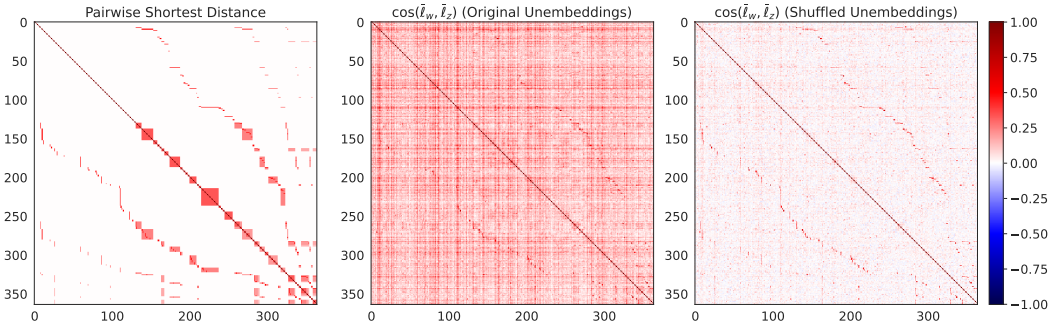

Figure 14: WordNet **verb** hierarchy is encoded in Gemma representation space.

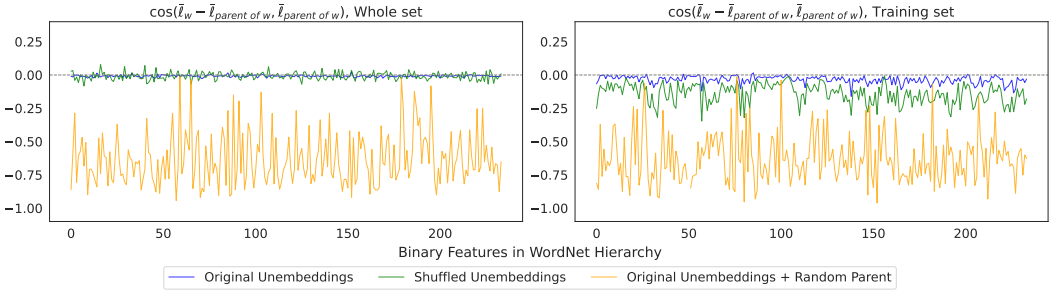

Figure 15: WordNet **verb** hierarchy is encoded as the orthogonal structure predicted by statement (a) in Theorem 8.

ent w. Then, the parent vector is $v_w = \frac{1}{N_a+N_b}(\sum a_i + \sum b_i)$, and the child vector is $v_z = \frac{1}{N_a}\sum a_i$. We know that $a_1, \ldots, a_{N_a}, b_1, \ldots, b_{N_b}$ are mutually orthogonal and $\|a_i\|^2 = \|b_i\|^2 = d$ with high probability. Therefore, we have

$$(v_z - v_w)^\top v_w = \tag{G.1}$$

$$= \left(\frac{N_b}{N_a(N_a+N_b)}\sum a_i - \frac{1}{N_a+N_b}\sum b_i\right)^\top \left(\frac{1}{N_a+N_b}\sum a_i + \frac{1}{N_a+N_b}\sum b_i\right) \tag{G.2}$$

$$= \frac{N_b}{N_a(N_a+N_b)^2}(\sum a_i)^\top \sum a_i - \frac{1}{(N_a+N_b)^2}(\sum b_i)^\top \sum b_i \tag{G.3}$$

$$= \frac{N_b}{N_a(N_a+N_b)^2}(N_a d) - \frac{1}{(N_a+N_b)^2}(N_b d) \tag{G.4}$$

$$= 0 \tag{G.5}$$

with high probability.

In words, the orthogonality between the child-parent vector and the parent vector can be derived from the set inclusion relationships. We can say that the set inclusion is the hierarchical relation as defined in Definition 2, but natural semantic hierarchy is our main focus in this paper. In this context, the right panel of Figure 5 shows that the violation of the set inclusion relationships makes the cosine similarity not close to 0, suggesting that semantic hierarchy is encoded as orthogonality.